Supplementary Information for

Development of experimental techniques for parameterization of multi-scale lithium-ion battery models

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S1 Electrode and cell dimensions

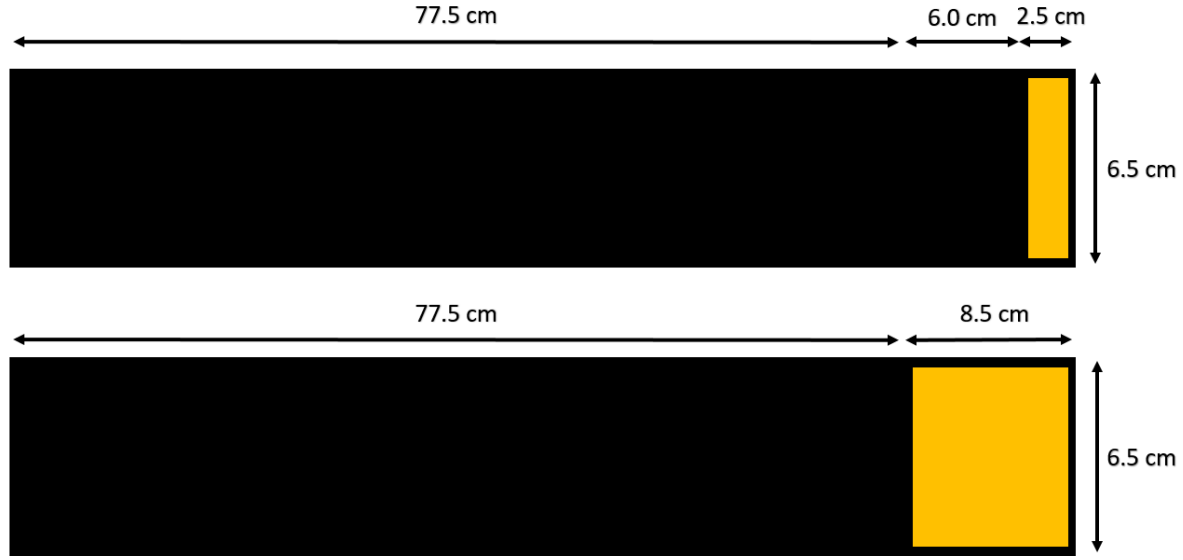


Figure S1. Dimensions for the asymmetric negative electrode coating with black denoting the graphite coating and orange denoting the bare current collector.

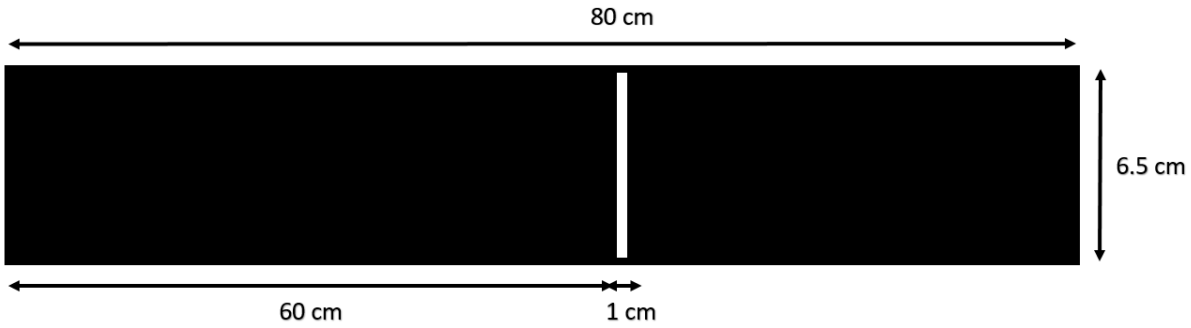


Figure S2. Dimensions for the asymmetric positive electrode coating with black denoting the NMC coating and white denoting the tab.

Item	Condition / Note	Specification
2.1 Energy	Std. charge / discharge	Nominal 18.20Wh Minimum 17.60Wh
2.2 Nominal Voltage	Average	3.63V
2.3 Nominal Shipping SOC		30%
2.4 Standard Charge (Refer to 4.2.1)	Constant current Constant voltage End current(Cut off)	0.3C (1,455mA) 4.2V 50mA
2.5 Max. Charge Voltage		4.20 ± 0.05V

Table S1. Data sheet from the manufacturer.

S2 SEM and EDS analysis of the positive electrode

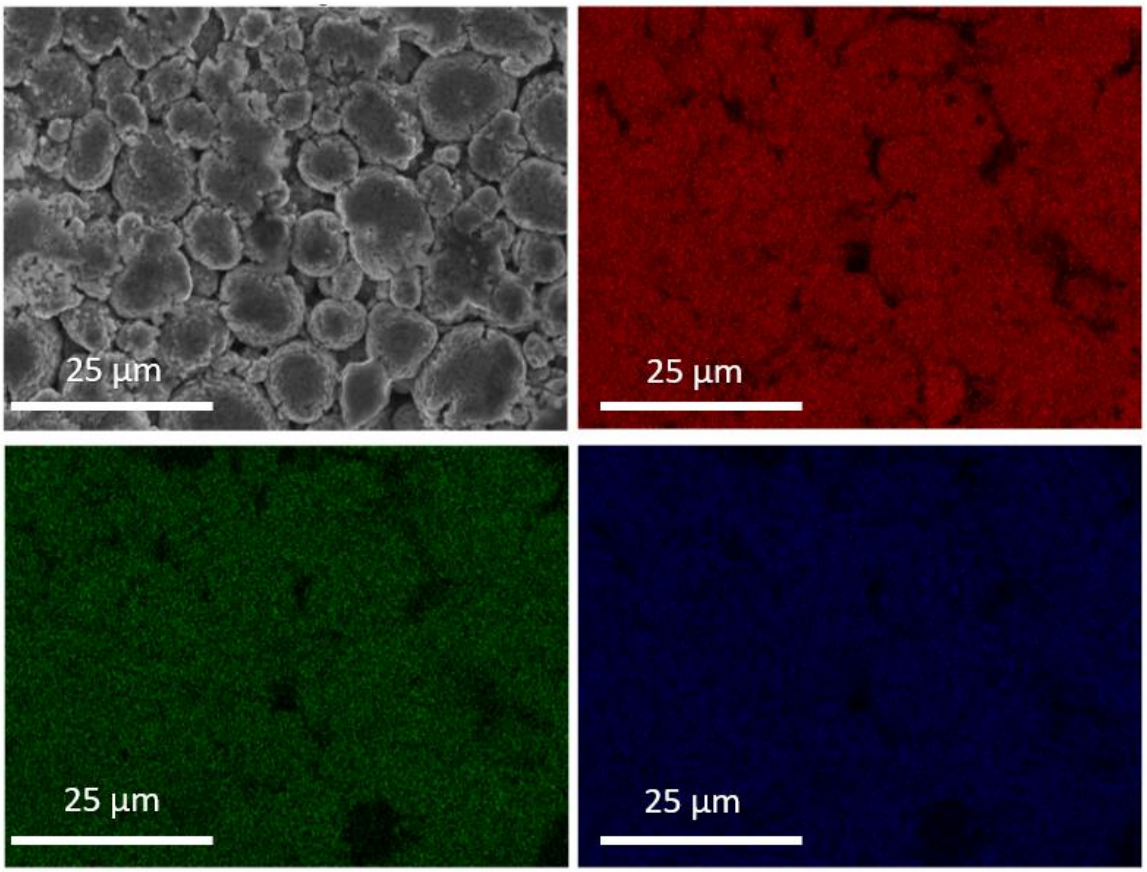


Figure S3. Top-view EDS images of the positive electrode extracted from M50 displaying nickel (red), cobalt (green), and manganese (blue) content.

EDS of Slice		1	2	3	Average
	Ni	60.3	61.1	60.7	82.9
	Co	7.9	7.6	7.75	10.6
	Mn	3.9	3.6	3.75	5.1
	Al	1.3	0.7	1	1.36

Table S2: EDS analysis of the FIB-SEM data

S3 SEM and EDS analysis of the negative electrode

S3.1 Average section EDS element analysis

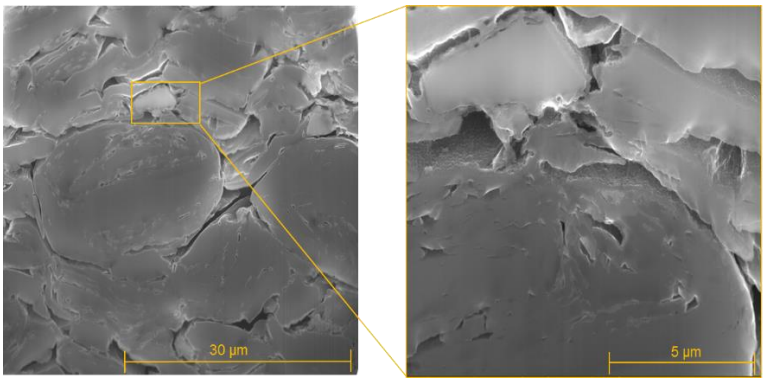
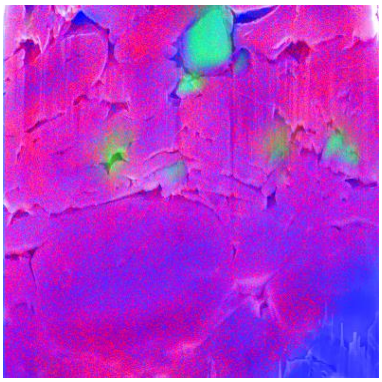


Figure S4. Cross section FIB-SEM images of the graphite-SiO_x electrode obtained from LG M50.

Element	Weight (%)	
	Area 1	Area 2
C	60.2	77.52
O	18.23	6.82
F	0.84	0.62
Si	7.1	5.78
P	0.35	0.24
S	1.04	0.88
Cu	12.24	8.14
Total	100	100

Table S3.1 EDS analysis of graphite-SiO_x electrode

S3.2 Average point EDS analysis



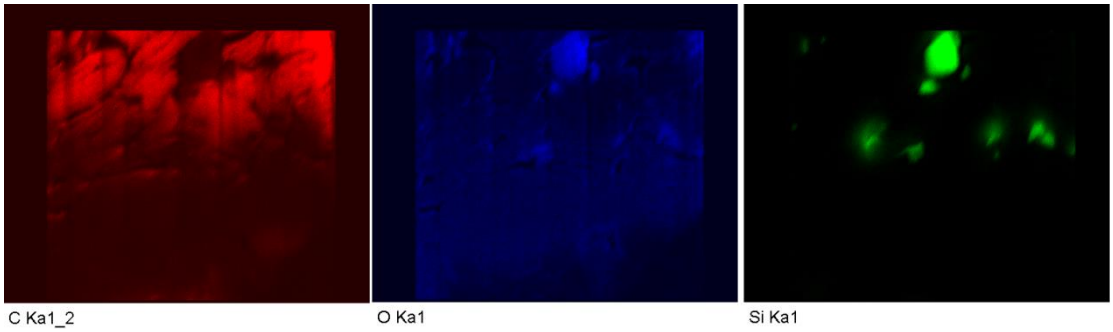


Figure S5. FIB-SEM and EDS images of the positive electrode extracted from M50 displaying carbon (red), oxygen (blue) and silicon (green) content.

	Weight %		
Element	Area 1	Area 2	Area 3
C	40.12	54.88	
O	31.92	18.69	36.47
Si	15.84	21.93	62.8
S	1.27	0.58	0.73
Cu	10.85	3.92	
Total	100	100	100

Table S3.2 EDS analysis of NMC electrode

S 3.5 ICP-OES analysis of the positive electrode

In the ICO-OES expriment, mixed standard solutions of 0.0, 2.0, 4.0, 6.0, 8.0 and 10.0 mg/L were prepared. The standard solution containing 22 elements, include Li, Ni, Mn, Al,B, Ba, Be, Bi, Cd, Co, Cr, Cu, Fe, Ga, K, Mg, Na, Pb, Se, Te, Tl and Zn.

Samples obtained from the electrode were dissolved, diluted and then analysed. Results were expressed as metal element concentrations by comparison with external standards. the weight percentage and corresponding atomic ratio of each element is shown in table S4..

Element	Atomic weight (g/mol)	Sample 1		Sample 2		Atomic ratio (average) where M=1
		w (%)	Atomic ratio	w(%)	Atomic ratio	
Li	6.9	8.79	8.07	9.45	8.63	8.24
Ni	58.7	74.03	7.99	74.13	7.96	7.87
Mn	54.9	4.91	0.57	4.91	0.56	0.55
Co	58.9	9.30	1.00	9.35	1.00	0.98
Al[a]	27.0	2.97	0.70	2.16	0.50	0.59

a：The contribution of aluminium from the aluminium disk is deducted.

Table S4. ICP-OES analysis of NMC electrode

S4 3D reconstruction of the electrodes

The 2D images of the electrodes were aligned and cropped accordingly: 62 □ 55 □ 57 µm^3 for the positive electrode and 47 □ 49 □ 20 µm^3 for the negative electrode, in order to remove any unwanted areas of the morphologies of the electrodes and reconstructed with the use of commercial software Synopsys Simpleware Scan IP [1]. A median filter was applied in order to remove noise.

The segmentation was achieved using a binary thresholding in order to reconstruct the 3D pore network. Then, the copy of pore domain was created and inverted on all slices in the whole cuboid domain by employing Boolean operation to create a new mask of solid constituents.

The volume fractions of the electrode components: active material and non-active, comprising of carbon black and pores were calculated by dividing the number of voxel assigned to one phase by the total number of voxels.

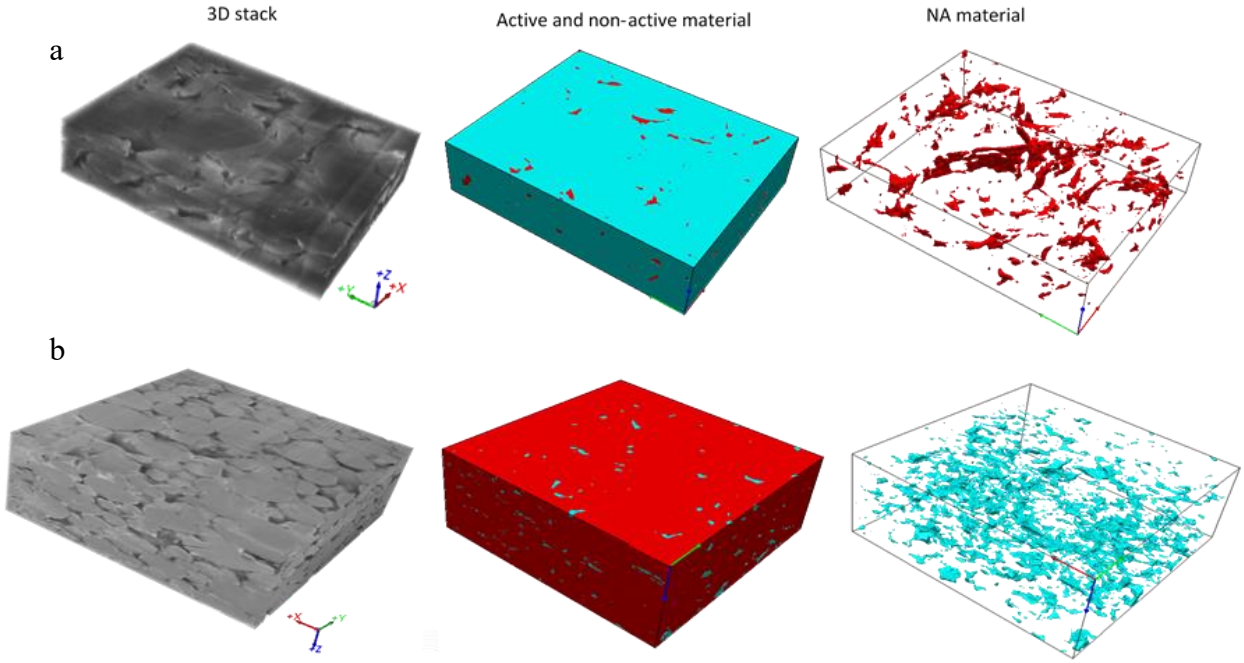


Figure S6. Reconstruction of M50 electrodes: negative electrode (top) and positive electrode (bottom) by Synopsys Simpleware Scan IP software.

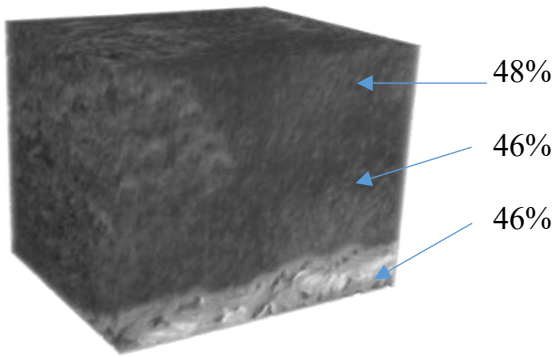


Figure S7. 3D reconstruction of ceramic coated separator and the calculated porosities

S5 Diffusion Coefficients

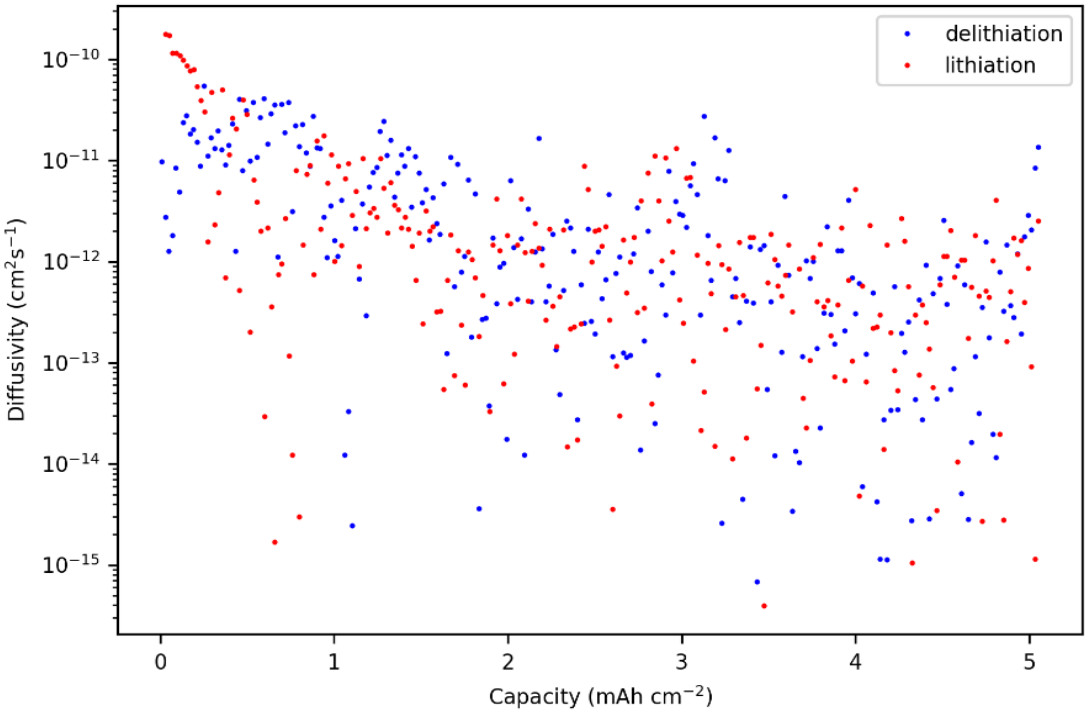


Fig S8 Diffusion coefficients calculated using the following equation modified from Weppner and Huggins. [42] (with no applied correction factor for particle size increase).

$$D_s = \frac{4}{\pi\tau}\left(\frac{m_{AM}V_M}{M_{AM}S}\right)^2\left(\frac{\Delta E_s}{\Delta E_t}\right)^2, \quad \left(\tau \ll \frac{R^2}{D_s}\right)$$

As is observed in S8 the voltage transients are slightly curved, and therefore care must be taken when fitting the SAND equation. The fits can be improved by utilizing part of the voltage transient curve. However the data obtained utilizing this methodology has significantly higher resolution than the data obtained from the titration measurements as shown in. S9.

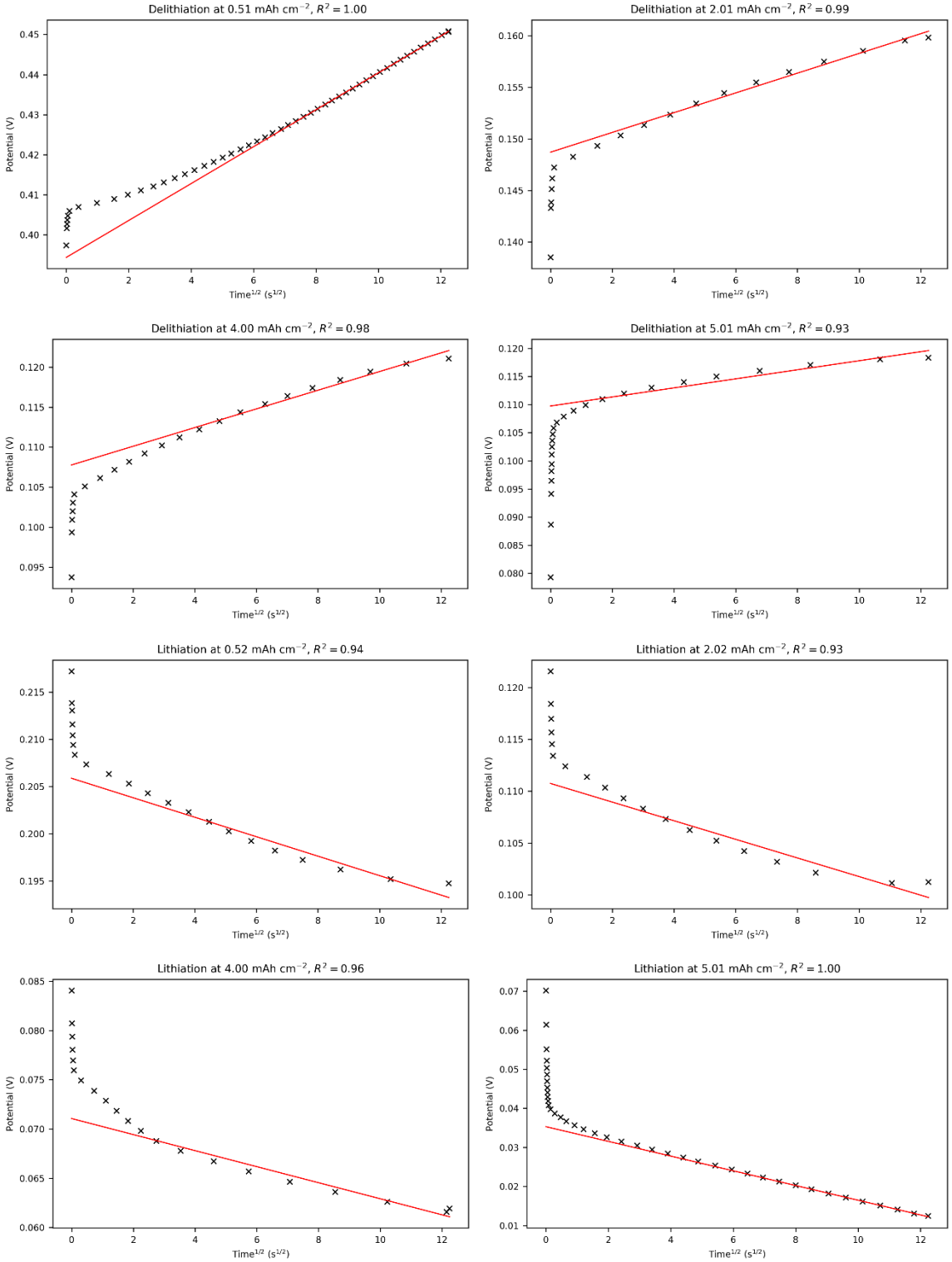


Fig S9 Voltage transient for lithiation and delithiation current pulses taken from 0.52, 2.02, 4 and 5.01 mAh cm^2 capacities, with the SAND equation fit for the negative electrode GITT ½ cell.

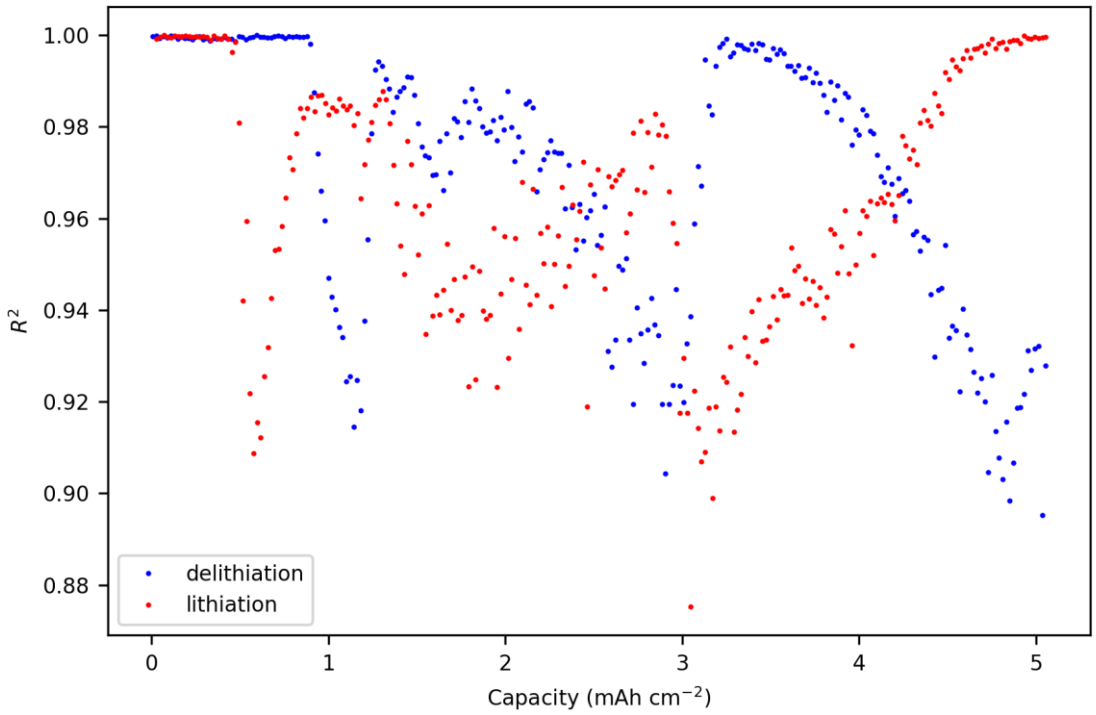


Fig S10 R^2 for SAND fit to voltage transient for lithiation and delithiation of the negative electrode (1/2 cell data)

S6 Electrochemical impedance spectroscopy measurement of half cells

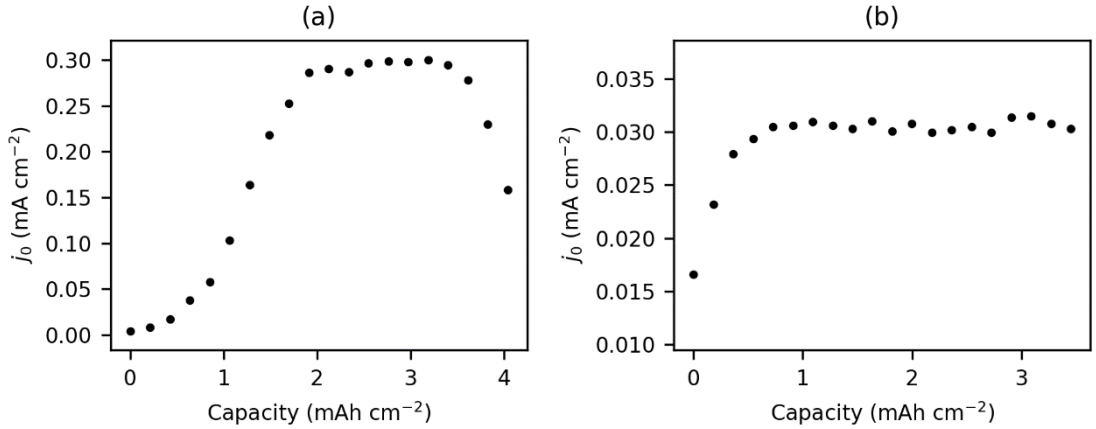


Figure S8. Exchange current densities calculated for (a) NMC and (b) graphite-SiO_x half cells at different states of charge at 25 □C.

References:

[1] www.simpleware.com, 2019.